

Weapon Detection and Classification in CCTV Surveillance: A Comparative Study of YOLO, RT-DETR and ResNet50

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Abstract—Weapon detection in surveillance imagery is an important task for public safety, since the amount of video footage is too large for manual monitoring. In this work we compare five deep learning models on the HARIS dataset, including four YOLO generations (YOLOv5nu, YOLOv8n, YOLO11n, YOLO26n), the transformer based detector RT-DETR-L, and a ResNet50 classifier. All models were trained using transfer learning with the same hyperparameters. A small experiment showed that full fine-tuning worked better than freezing layers, so we used that approach for all main runs. The four YOLO nano models reached similar mAP50 scores between 83.4% and 84.4%, while RT-DETR-L scored lower even though it has around 15 times more parameters. YOLO26n reached its best performance in fewer epochs than the older versions. ResNet50 performed better than the YOLO models on pure image classification with 91.7% accuracy. Looking at the per-class results, the biggest source of errors is confusing weapons with background rather than confusing guns with knives. Overall, the newer YOLO versions did not outperform the older ones, and a two-proportion z-test confirmed that dedicated classifiers worked significantly better than detectors on the classification task ($p < 0.001$).

Index Terms—CCTV surveillance, object detection, ResNet, transfer learning, weapon detection, YOLO

I. BACKGROUND AND RELATED WORK

The increasing use of surveillance cameras in public spaces has created a growing need for automated threat detection systems. Since modern camera networks generate large amounts of video, manual monitoring is often impractical. Automated weapon detection using computer vision can help identifying potential threats more quickly, but the task remains challenging because of varying image quality and labeled data.

Recent advances in deep learning have made object detection systems more accurate and fast enough for real-time use. Among these approaches, the YOLO family of models have become a popular choice due to its balance between detection performance and inference speed. Bhatti et al. [1], compared several deep learning models for weapon detection in CCTV footage and found YOLO to provide one of the best trade-offs between speed and accuracy. Similar results were reported by León et al. [2], who applied a YOLOv5-based system for firearm detection in educational systems.

Transformer-based detection models have emerged as alternative to traditional CNN-based approaches. RT-DETR, introduced by Zhao et al. [3], was designed as a real-time transformer detector and achieved competitive results against YOLO on standard object detection benchmarks.

In this project we compare four YOLO generations (YOLOv5nu, YOLOv8n, YOLO11n and YOLO26n) on the HARIS weapon detection dataset [4]. We also compare the results to the published benchmark values of RT-DETR-L, and evaluate a ResNet50 classifier on the classification task.

II. DATA EXPLORATION

The HARIS dataset (*Hostile Activity Recognition and Intelligent Surveillance*) is a publicly available dataset hosted on Kaggle. The images are primarily collected from CCTV and surveillance-style footage, making it representative of real-world deployment conditions. The dataset contains approximately 9,700 training images and 1,800 validation images with two object classes: *gun* and *knife*. The class distribution is not perfectly balanced, with guns occurring more frequently than knives, but it is relevant for this study because it reflects surveillance scenarios. The dataset also contains subsets of close-up weapon images, as well as images containing no weapons, increasing the diversity of the dataset.

III. METHOD DESCRIPTION

For object detection, four YOLO models were trained: YOLOv5nu, YOLOv8n, YOLO11n and YOLO26n. All experiments were conducted using the Ultralytics framework, with model training on an NVIDIA A100 GPU. Detection models were initialized using weights from images pretrained on COCO dataset.

These models were selected to represent multiple generations of the YOLO family in comparable model sizes. The nano variants were chosen because they are more suitable for a moderately sized dataset, and larger models require more computational resources. RT-DETR-L was not retrained in this study, the published benchmark values provided with the HARIS dataset were used for comparison.

For image classification, a binary ResNet50 classifier pre-trained on images from ImageNet was trained and fine-tuned to predict whether an image contains a weapon or not.

A. Experiment Design

All detection models were trained using a batch size of 64, an input resolution of 640×640 pixels, and a maximum of 150 epochs. The best checkpoint was selected based on the mAP50-95 metric. Apart from model-specific defaults, the same training settings were used across all experiments. Before the main experiments, a small study was conducted on layer freezing during transfer learning.

Three configurations were compared: full fine-tuning, freezing the first 10 layers, and freezing the first 20 layers. Based on the results, full fine-tuning was used in all subsequent experiments.

For the classification task, we used a ResNet50 model pretrained on ImageNet, imported from the Keras library. Images were resized to 224×224 pixels, and followed a custom classification head with batch normalization, global average pooling, 40 % dropout, and a sigmoid output layer for binary classification. The last three layers were fully fine-tuned, based on a similar transfer learning test.

IV. EVALUATION

A. Performance Metrics

The object detection models were evaluated using precision, recall, mAP50, and mAP50-95. Precision measures how often a predicted object is correct, while recall measures how many true objects are successfully detected. Both are important in a security setting, where missed detections and false alarms can have practical consequences.

mAP50 and mAP50-95 are standard object detection metrics that summarize detection quality across different intersection over union (IoU) thresholds, in the range 0.5 to 0.95.

B. Statistical Testing

To check whether the differences in classification accuracy between models are statistically meaningful, we used a two-proportion z-test, which compares the proportion of correct predictions from two models on the same validation set. A p-value below 0.05 indicates that the difference is statistically significant at the 95% confidence level. We applied this test to two pairwise comparisons: ResNet50 against YOLO26n, and YOLO26n against YOLOv5nu. We also report the 95% confidence interval for the accuracy difference in each comparison.

V. RESULTS AND ANALYSIS

A. Freeze Layer Experiment

The results in Fig. 1 shows that full fine-tuning gave the highest mAP50-95 score of 52.3%, while freezing 10 layers reached 51.0%, only slightly lower. Freezing 20 layers caused a much bigger drop to 37.1%, around 29% lower than full fine-tuning. The small difference between freezing 0 and 10 layers shows that the early features from COCO transfer fairly well to HARIS, while the bigger drop at 20 layers suggests

that the later layers need to adapt more to the CCTV style images. Based on this, we used full fine-tuning for the rest of our experiments.

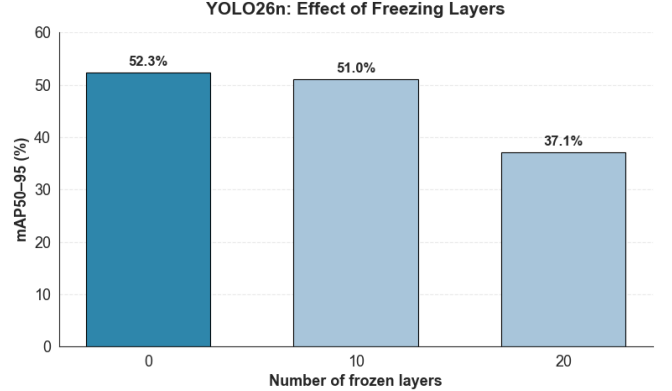


Figure 1. YOLO26n freeze-layer experiment comparing full fine-tuning with 10 and 20 frozen layers.

B. Object Detection Performance

The four YOLO models reach very similar mAP50 scores between 83.4% and 84.4%, so all generations seem about equally good at finding weapons, as shown in Table I. The differences are within 1.0 percentage points, which is likely within the noise we get from single training runs. This is in line with Bhatti et al. [1], who found that YOLO is a reliable choice for CCTV weapon detection.

Table I
OBJECT DETECTION RESULTS ON THE HARIS VALIDATION SET. BOLD VALUES INDICATE THE BEST PERFORMANCE PER METRIC.

Model	P	R	mAP50	mAP50-95	Ep.
YOLOv5nu	85.3%	80.0%	83.9%	52.1%	144
YOLOv8n	86.1%	79.0%	84.4%	52.3%	124
YOLO11n	84.4%	80.4%	84.2%	53.6%	148
YOLO26n	86.6%	77.1%	83.4%	53.4%	98
RT-DETR-L	87.2%	70.4%	78.0%	47.6%	—

The differences are larger on mAP50-95, which measures bounding box tightness across multiple IoU thresholds. YOLO11n and YOLO26n score highest (53.6% and 53.4%), slightly ahead of YOLOv5nu and YOLOv8n (52.1–52.3%). So the newer YOLO models draw a bit tighter boxes around the weapons, even though they detect about the same number of them.

RT-DETR-L behaves differently. It has the highest precision (87.2%) but the lowest recall (70.4%), which gives lower overall mAP scores (mAP50 = 78.0%, mAP50-95 = 47.6%). It is more careful with its predictions. It predicts less, but is more often right when it does. Even so, RT-DETR-L has about 15 times more parameters than the YOLO nano models (around 32M compared to 2.5M) and still does not perform better on this dataset. This is different from what Zhao et al. [3] showed, where RT-DETR beat YOLO on COCO. Their results don't

seem to carry over to a smaller and more specialized dataset like ours.

Convergence behavior: The Epoch column in Table I shows when each model reached its best mAP50 score. YOLO26n reached its best at epoch 98, much earlier than the other YOLO variants which peaked between epochs 124 and 148. So even though the final accuracy is similar, YOLO26n got there faster, which suggests better training efficiency. These results are from single training runs per model, so they should be seen as indicative rather than statistically certain.

C. Per-Class Analysis

Fig. 2 shows the confusion matrix for YOLO26n on the validation set, using the default confidence threshold of 0.25. The matrix lets us see what kind of errors the model makes, more than what we get from the mAP scores alone. The model rarely confuses guns and knives with each other (only 1 gun was predicted as a knife, and 6 knives as guns). Most errors are with the background class instead. Out of 886 gun instances, 222 (25.1%) were missed and classified as background, while 24 out of 322 knives (7.5%) were missed the same way. The model also made 159 false-positive gun detections and 49 false-positive knife detections on background regions.

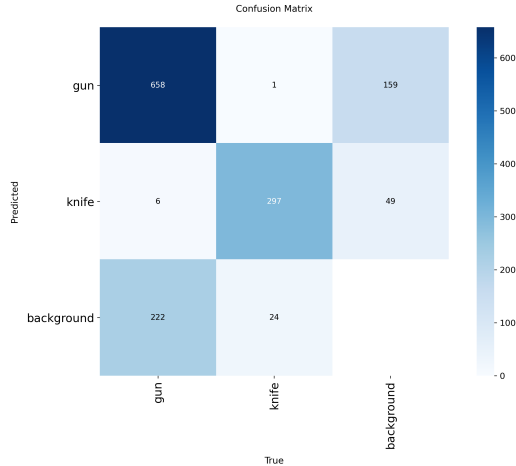


Figure 2. Confusion matrix for YOLO26n on the HARIS validation set at confidence threshold 0.25. Rows are predicted classes and columns are ground truth.

Knife detection was actually more accurate than gun detection. The per-class metrics give 84.4% precision and 92.2% recall for knives ($F1 = 0.881$), compared to 80.4% and 74.3% for guns ($F1 = 0.772$). This was a bit unexpected since knives are usually thought to be harder because of their thin shape. One possible reason is that the knife images in HARIS often show the knife clearly in the foreground, while the gun images have more varied angles and partial occlusion. Overall, the biggest source of error is not telling gun and knife apart but telling weapons apart from background.



Figure 3. YOLO26n detection of guns with confidence 0.5 and 0.9.

D. Example Detection Results on Validation Images

To illustrate how the detector behaves in practice, Fig. 3 shows example predictions produced by YOLO26n on validation images. The model outputs a bounding box around the detected object together with a confidence score representing the estimated probability that the object belongs to the predicted class. In both examples, the detector successfully localizes the weapons, with different confident scores (0.5 and 0.9), where the lower score corresponds to a more challenging image.

E. Classification Comparison

We also looked at how the models do on plain binary classification. ResNet50 was trained directly for this, while for the YOLO models we converted their detections into per-image labels by counting TP, TN, FP, and FN values from each model’s confusion matrix at confidence threshold 0.25. The two setups are not fully comparable since ResNet predicts once per image while YOLO works per detection, but the comparison still gives a useful overview. The comparison should therefore be interpreted as a practical comparison of weapon-presence recognition rather than a direct comparison of model architecture. Table II summarizes the results.

Table II
CLASSIFICATION RESULTS AT CONFIDENCE THRESHOLD 0.25. RESNET50 METRICS ARE COMPUTED PER IMAGE, WHILE YOLO METRICS ARE DERIVED FROM PER-INSTANCE CONFUSION MATRICES. BOLD VALUES INDICATE THE BEST PERFORMANCE PER METRIC.

Model	Precision	Recall	Accuracy
YOLOv5nu	76.9%	81.7%	71.2%
YOLOv8n	77.9%	82.4%	72.6%
YOLO11n	77.7%	82.6%	72.4%
YOLO26n	79.6%	82.2%	74.7%
ResNet50	89.4%	98.1%	91.7%

ResNet50 is clearly the best on this task with 91.7% accuracy, compared to about 71 to 75% for the YOLO variants. This is not surprising since ResNet50 is built for image classification while YOLO is built for object detection. When we reduce YOLO to a yes/no classification we lose some of the information that detection would normally use.

Among the YOLO models, YOLO26n is the best with 74.7% accuracy, which fits what we saw in the detection results. ResNet50 has very high recall (98.1%) so it rarely misses a weapon, but the lower precision (89.4%) means around one in ten positive predictions is a false alarm. This is partly because of the class imbalance in the training data, where about 75% of the images contain weapons. To check

whether the differences are statistically significant, we ran two-proportion z-tests on the classification results. Table III summarizes them.

Table III
TWO-PROPORTION Z-TEST RESULTS FOR PAIRWISE CLASSIFICATION ACCURACY COMPARISONS ON THE HARIS VALIDATION SET (N = 1801 PER MODEL).

Comparison	Δ Acc.	z	p	95% CI
ResNet50 vs YOLO26n	17.0%	13.63	<0.001	[14.6, 19.4]%
YOLO26n vs YOLOv5nu	3.5%	2.36	0.018	[0.6, 6.4]%

The difference between ResNet50 and YOLO26n is highly significant ($p < 0.001$), confirming that the 17 percentage point gap is not due to chance. The smaller difference between YOLO26n and YOLOv5nu is also significant ($p = 0.018$), but the effect size is much smaller and the confidence interval is much wider.

VI. DISCUSSION

The results suggest that newer YOLO versions do not automatically perform better on a smaller and more specialized dataset such as HARIS. The differences between the YOLO models were small, which may be because all models were already strong enough for the dataset or because single training runs do not capture the full variation between models.

RT-DETR-L performed worse than expected compared to the YOLO models. This differs from results on larger benchmark datasets, where RT-DETR has shown strong performance. One possible explanation is that transformer-based detectors may require more data to adapt well, while the lightweight YOLO models are easier to fine-tune on a specialized weapon dataset.

The per-class analysis showed that the main problem was not confusing guns with knives, but confusing weapons with background. This is important for real surveillance systems because missed weapons are dangerous, while too many false alarms can make the system less useful in practice.

The classification comparison also shows that detection and classification should not be treated as the same task. ResNet50 was better for image-level weapon recognition, but it cannot show where the weapon is located. A practical system could therefore combine a classifier for image-level screening with an object detector for localization.

VII. CONCLUSION

This report compared YOLOv5nu, YOLOv8n, YOLO11n, YOLO26n, RT-DETR-L, and ResNet50 on the HARIS weapon detection dataset. The YOLO nano models achieved very similar detection performance, with mAP50 scores around 84%. YOLO11n and YOLO26n reached slightly higher mAP50-95 scores, while YOLO26n reached its best checkpoint in fewer epochs.

RT-DETR-L did not outperform the YOLO models, despite having more parameters. It achieved high precision but lower

recall, which made its overall detection performance weaker on this dataset.

For image-level classification, ResNet50 performed clearly better than the YOLO models, reaching 91.7% accuracy. The statistical test confirmed that the difference between ResNet50 and YOLO26n was significant. However, ResNet50 only predicts whether a weapon is present, while YOLO also localizes the weapon. Therefore, the best model choice depends on whether the goal is classification, localization, or both.

Future Work

A natural next step would be a hybrid setup that uses ResNet50 for image-level classification and YOLO for localization when a weapon is detected. This combines the strengths of both and could reduce false alarms.

A larger and more varied dataset would also help, especially with harder background images, since the per-class analysis showed that confusing weapons with background is the main source of errors. The HARIS dataset is fairly limited in size and contains mostly clear images, while real CCTV footage is often lower quality and has more unusual angles and occlusion.

Finally, it would be interesting to extend this to video footage, since most surveillance systems work with video and not single images. Looking across frames could help reduce false alarms, since a real weapon should show up consistently while a false detection usually does not.

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